TEZPUR UNIVERSITY

Evaluating Soil Strength Parameters and Bearing Capacity for Field Applications (NMH)

"An In-depth Analysis of Friction Angle, Unit Weight, Cohesion, and Failure Mechanisms for Soil Bearing Capacity Estimation"

GROUP-E

GEOTECHNICAL ENGINEERING-II

Dr. Utpal Kumar Das

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This document includes the details of all members who contributed to this assignment.

Name	Roll No.	Course & Year
Ram Kumar Panjiyara	CEB22049	B.Tech 3rd Year, Civil Engineering
Vikrant Verma	CEB22053	B.Tech 3rd Year, Civil Engineering
Md Firdoush	CEB22060	B.Tech 3rd Year, Civil Engineering
Tadar Nyikum Abel	CEB22086	B.Tech 3rd Year, Civil Engineering
Biman Pangia	CEB22088	B.Tech 3rd Year, Civil Engineering

To determine the bearing capacity of soil, it is essential to obtain the following soil properties:

1. **Soil Type**
   - **Test Required**: The soil type is determined by performing Atterberg Limits tests, specifically the Liquid Limit and Plastic Limit tests. The Casagrande method is used to assess these limits, which helps categorize the soil type based on its plasticity and behavior under varying moisture conditions.
2. **Unit Weight of Soil (γ)**
   - **Test Required**: The unit weight of soil can be evaluated using the Core Cutter Method. This method involves extracting a known volume of soil and measuring its weight, providing a straightforward calculation of the unit weight, which is critical in bearing capacity calculations.
3. **Cohesion Value (C)**
   - **Test Required**: Cohesion is determined by conducting either the Direct Shear Test or the Unconfined Compressive Strength Test. These tests help establish the cohesive strength of the soil, which contributes significantly to its bearing capacity, particularly in cohesive soils.
4. **Internal Friction Angle (Φ)**
   - **Test Required**: The Internal Friction Angle is found through the Direct Shear Test. This parameter reflects the soil's resistance to sliding along internal surfaces and is crucial in understanding soil stability under load.

A. Lets perform first core cutter method:

REQUIRED APPARATUS for core cutter method:

i. cylindrical core cutter: A 100 mm internal diameter and 130 mm long cutter ii. Steel rammer: A 9 kg rammer with a total length of about 900 mm iii. Steel dolly: A 100 mm internal diameter and 25 mm high dolly iv. Weighing balance: A balance with an accuracy of 1 g

v. Palette knife: A knife for trimming the soil vi.

Straight edge or steel rule: A tool for measuring

vii. Spade or pickaxe: A tool for digging

viii. Container ix. Oven

principle:

The core cutter method is used to determine the in-situ bulk density or unit weight of soil. It involves driving a cylindrical core cutter into the soil and extracting a sample to measure its volume and mass. By calculating the mass of the soil sample and knowing the cutter's volume, the bulk density is determined. This method is generally suitable for cohesive soils and is not ideal for coarse-grained.

Procedure:

i. We cleared and leveled the surface where the test was to be conducted.

ii. We positioned the core cutter (with known dimensions) vertically on the soil surface.

iii. Using a rammer, we drove the core cutter into the soil until it was fully embedded, ensuring it remained upright.

iv. We carefully dug around the cutter to remove it without disturbing the soil inside.

v. Any soil protruding from the cutter was trimmed off with a straightedge.

vi. We weighed the cutter filled with soil and recorded the mass. vii. We weighed the empty cutter and recorded this value. viii. Using the mass difference and the known volume of the cutter, we calculated the bulk density.

Calculation :-

For core cutter:

Volume of mould = $\pi r^2 h$

$= 3.14 \times 5 \times 5 \times 13$

$= 1021 \text{ cm}^3$

Mass of mould = 967 gm

Weight of sample + mould = 2776 gm

Mass of soil = 1809 gm

Bulk density = 1.772 gm/cm^3

For dry density/moisture content:

Mass of container empty = 13.65 gm

Container + mass of wet soil = 64.64 gm

Container + mass of dry soil = 56.390 gm

Mass of water/Mass of soil = Moisture content = (64.64 – 56.390)/(56.390 – 13.65) x 100%

w = 19.303 %

Dry density = bulk density / (1 + w) = 1.485 gm/cm^3

Now unit weight for given sample:

(γ) = 17.365 KN/m^3

B. Lets perform the Liquid Limit Test:

REQUIRED APPARATUS for Liquid Limit Test:-

i. Casagrande apparatus.

Principle-:

The Casagrande method measures the liquid limit, which is the moisture content at which soil begins to flow under specific conditions. This test involves placing a soil paste in a Casagrande cup and creating a groove in the soil. The cup is dropped repeatedly from a standard height, and the moisture content at which the groove closes over a specific number of blows (usually 25) is the liquid limit. This limit indicates the boundary between plastic and liquid behavior of soil.

Procedure:-

i. We took a soil sample passing through a No. 40 sieve (425 µm).

ii.Water was gradually added to the soil to form a paste.

iii. We placed the soil paste in the Casagrande cup and smoothed the surface.

iv. Using the grooving tool, we created a groove in the soil paste from the back to the front of the cup.

v. The handle was rotated to lift and drop the cup at a rate of 2 drops per second.

vi. We recorded the number of blows required for the groove to close over a distance of 13 mm.

vii.We adjusted the moisture content and repeated the test until closure occurred at around 25 blows. viii. A sample of the soil paste that closed at 25 blows was taken to determine its moisture content.

C. Lets perform the Plastic Limit Test:

REQUIRED APPARATUS for Plastic Limit Test:-

- **Glass plate**: A 10 mm thick glass plate that is 45 cm square or larger
- **Porcelain evaporating dish**: A porcelain dish that is about 12 cm in diameter
- **Brass rod**: A brass rod that is 3 mm in diameter and 10 cm long
- **Hot air oven**: A thermostatically controlled oven with a non-corroding interior that maintains a temperature of around 1050°C to 1100°C
- **Balance**: A balance that is accurate to 0.01 g
- **Spatula**: A flexible spatula
- **Moisture containers**: Six moisture containers
- **Wash bottle**: A wash bottle containing distilled water
- **Glazed glass desiccators**: Glazed glass desiccators
- **425-micron sieve**: A 425-micron sieve

Principle:-

The plastic limit is the moisture content at which the soil begins to exhibit plastic behavior (i.e., can be molded without cracking). In this test, the soil is rolled into threads until it begins to crumble at a specific diameter (typically 3 mm). The moisture content at this point is the plastic limit. This value is part of the Atterberg limits and helps in classifying soil consistency and plasticity.

Procedure:-

i. We took a soil sample passing through a No. 40 sieve (425 µm).

ii. We added water to the soil until it became plastic enough to be molded.

iii. A small portion of soil was rolled on a glass plate to form threads approximately 3 mm in diameter.

iv. We continued rolling until the soil thread began to crumble at about 3 mm thickness.

v. The crumbled soil was collected in a container for moisture content determination.

vi. The process was repeated 2-3 times to ensure accuracy.

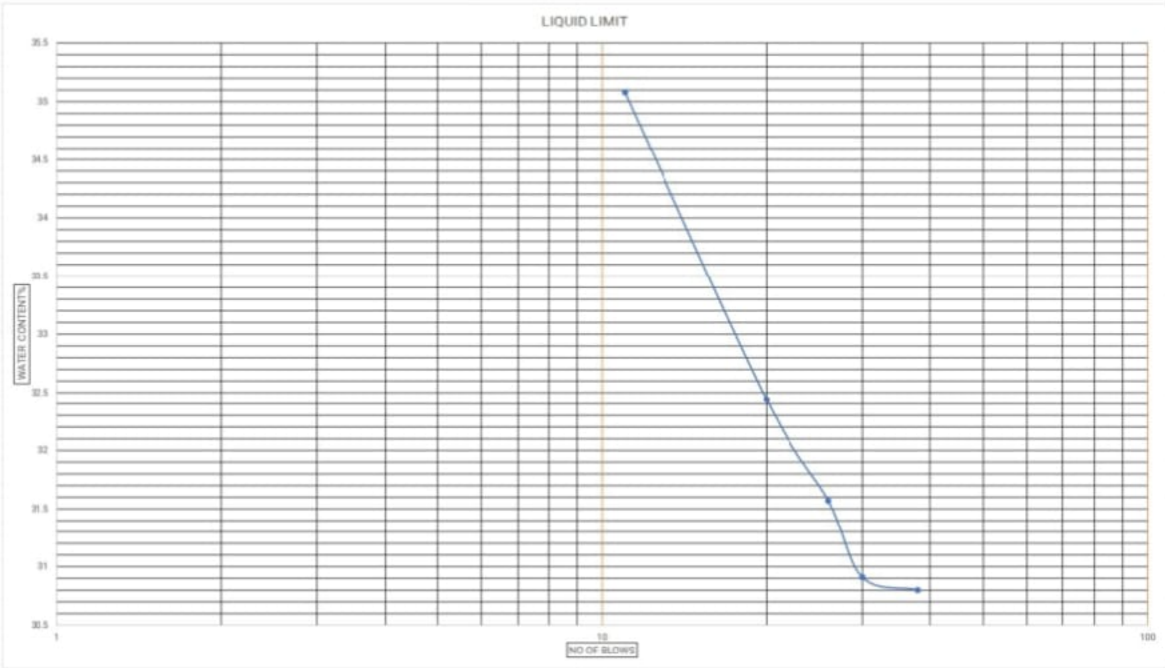
vii. The moisture content at which the soil crumbled at 3 mm thickness was recorded as the plastic limit.

Calculation for Liquid and Plastic limit test:-

From graph, water content corresponding 25 blows:

OBSERVATION TABLE

SL. NO.	NO. OF BLOWS	WATER CONTENT %
1	38	30.8
2	30	30.91
3	26	31.44
4	20	32.44
5	11	35.08



So, Liquid Limit = 32.95 %

And, from experiment,

Plastic limit = 19.04 %

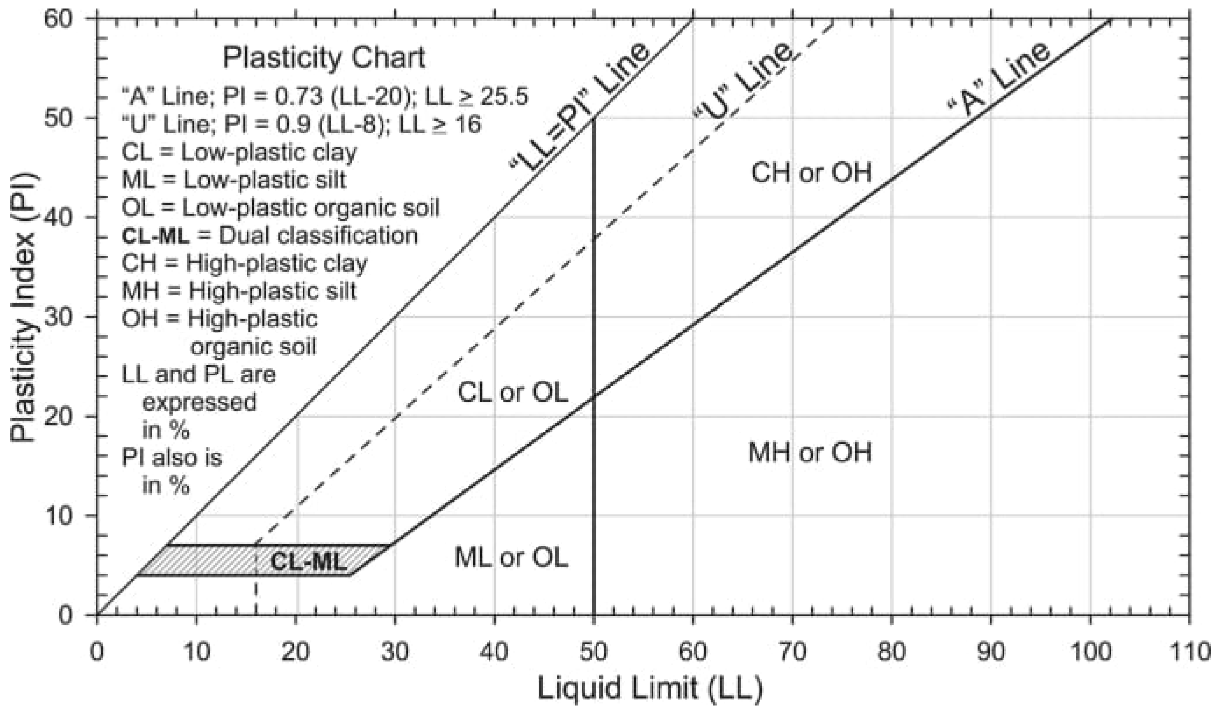
Now, I_p = 32.95 – 20

= 13.91 %

Value on A line correspond 32.95% w

= 0.73 x (32.95 - 20)

= 9.45



So, our soil is upper side on A line.

From atterburg curve, Our soil type is CL or OL

From observation our soil in - low plastic organic clay

D. Lets perform the Unconfined compressive strength test :-

REQUIRED APPARATUS for Unconfined compressive strength test :-

- **Loading device**: A device that applies the load at a consistent rate until the test is complete. This can be a hydraulic loading device, a screw jack with a proving ring, or a platform weighing scale with a screw-jack activated yoke.
- **Platens**: Two steel platens that transfer the axial stress from the loading device to the specimen. The platens should have a minimum Rockwell Hardness of 58 and a diameter that is at least equal to the sample's diameter.
- **Strain measurement devices**: Devices that measure the axial and lateral deformations, such as Linear Variable Differential Transformers (LVDTs), Compressometers, or Electrical Resistance Strain Gages.
- **Proving ring**: A proving ring with a sensitivity of 0.01 kg for soft soils and 0.05 kg for stiff soils.

- **Dial gauge**: A dial gauge with a least count of 0.01 mm.
- **Vernier calipers**: Vernier calipers with a least count of 0.1 mm.
- **Oven**: A thermostatically controlled oven with a non-corroding interior that can measure 110° ± 5° C.
- **Weighing balances**: Weighing balances with a least count of 0.01 g if the specimen weight is less than 100 g or 0.1 g if the specimen weight is more than 100 g.

Principle:-

The unconfined compressive strength (UCS) test is performed on cohesive soils to measure their compressive strength without any lateral confining pressure. A cylindrical soil sample is subjected to increasing axial stress until failure occurs. The maximum stress at failure is the UCS, which is used to determine soil strength and is particularly important in foundation and stability analysis for cohesive soils.

Procedure:-

i. We trimmed a cylindrical soil sample (typically 3.5 cm diameter and 7 cm length) with minimal disturbance.

ii. The sample's initial dimensions were recorded for accurate area calculation.

iii. We mounted the sample vertically in the unconfined compression testing machine.

iv. An axial load was gradually applied to the sample without any lateral confinement.

v. We measured the axial deformation and the corresponding load at regular intervals until failure.

vi. The maximum load applied before the soil failed was identified as the failure load.

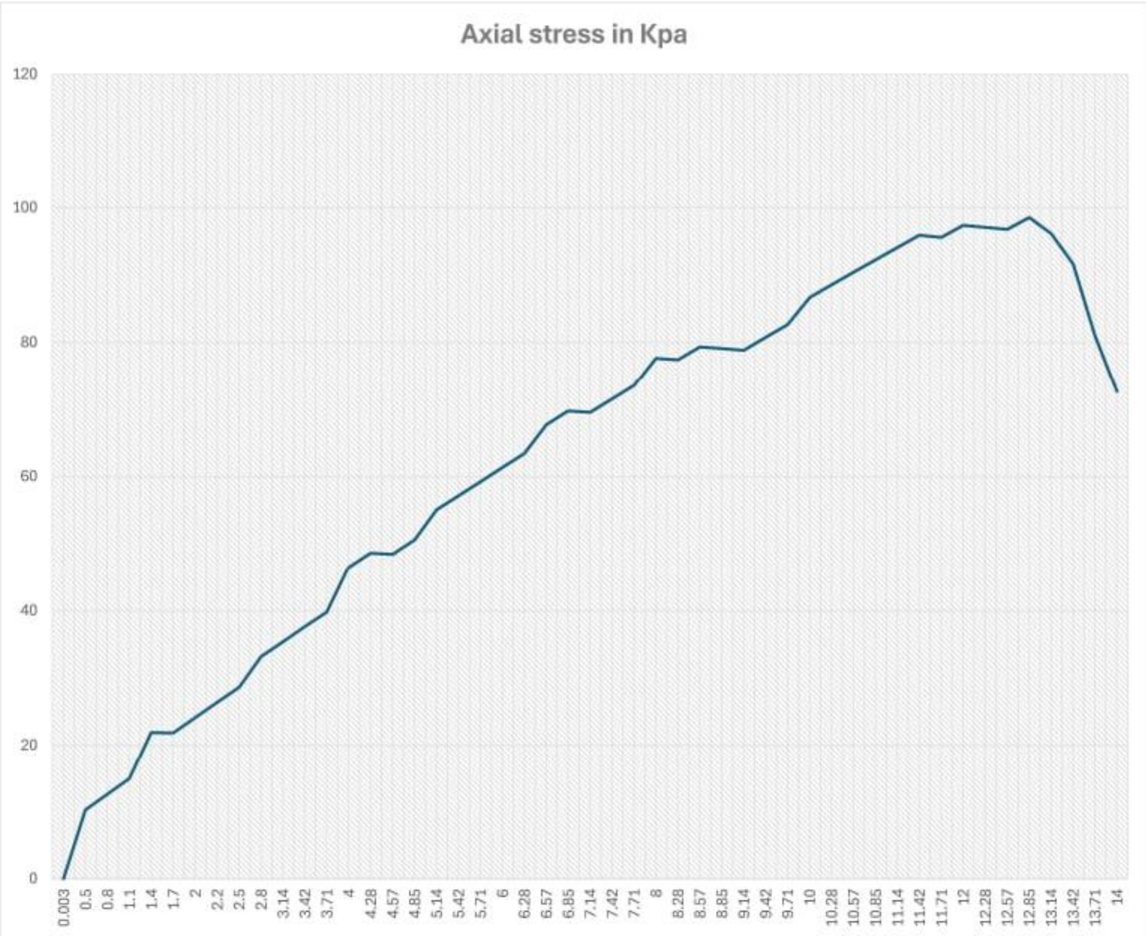
vii. We calculated the unconfined compressive strength using the failure load and cross-sectional area of the sample

Calculation:-

From Unconfined compressive strength test (UCS):

Dailgauge reading (in division)	Proving ring reading (in division)	Displacement in mm	Load KN	Axial starin %	corrected Area	Axial Stress in Kpa
20	0	0.2	0	0.003	965.007	0
40	3	0.4	0.0099	0.5	966.947	10.23841017
60	4	0.6	0.0122	0.8	969.871	12.57899246
80	5	0.8	0.0145	1.1	972.831	14.90495266
100	8	1	0.0214	1.4	975.773	21.93133034
120	8	1.2	0.0214	1.7	978.751	21.8646009
140	9	1.4	0.0237	2	981.747	24.14063909
160	10	1.6	0.026	2.2	983.755	26.4293447
180	11	1.8	0.0283	2.5	986.782	28.67908008
200	13	2	0.0329	2.8	989.827	33.23813151
220	14	2.2	0.0352	3.14	993.301	35.43739511
240	15	2.4	0.0375	3.42	996.182	37.64372374
260	16	2.6	0.0398	3.71	999.182	39.83258305
280	19	2.8	0.0467	4	1002.2	46.59748553
300	20	3	0.049	4.28	1005.132	48.74981594
320	20	3.2	0.049	4.57	1008.186	48.60214286
340	21	3.4	0.0513	4.85	1011.153	50.73416189
360	23	3.6	0.0559	5.14	1014.244	55.11494276
380	24	3.8	0.0582	5.42	1017.247	57.21324319
400	25	4	0.0605	5.71	1020.376	59.29186888
420	26	4.2	0.0628	6	1023.524	61.35664625
440	27	4.4	0.0651	6.28	1026.582	63.41432053
460	29	4.6	0.0697	6.57	1029.768	67.6851485
480	30	4.8	0.072	6.85	1032.863	69.70914826
500	30	5	0.072	7.14	1036.089	69.49209962
520	31	5.2	0.0743	7.42	1039.089	71.50494327
540	32	5.4	0.0766	7.71	1042.488	73.47806402
560	34	5.6	0.0812	8	1045.774	77.64583935
580	34	5.8	0.0812	8.28	1048.967	77.40948953
600	35	6	0.0835	8.57	1052.294	79.35044769
620	35	6.2	0.0835	8.85	1055.526	79.10747817
640	35	6.4	0.0835	9.14	1058.895	78.85578835
660	36	6.6	0.0858	9.42	1062.169	80.77810593
680	37	6.8	0.0881	9.71	1065.586	82.67751266
700	39	7	0.0927	10	1069.014	86.71542187
720	40	7.2	0.095	10.28	1072.35	88.59047885
740	41	7.4	0.0973	10.57	1075.827	90.44205063
760	42	7.6	0.0996	10.85	1079.206	92.29007252
780	43	7.8	0.1019	11.14	1082.728	94.11412654
800	44	8	0.1042	11.42	1086.151	95.93509558
820	44	8.2	0.1042	11.71	1089.718	95.62106894

840	45	8.4	0.1065	12	1093.309	97.41070457
860	45	8.6	0.1065	12.28	1096.799	97.10074499
880	45	8.8	0.1065	12.57	1100.487	96.77533674
900	46	9	0.1088	12.85	1103.973	98.5531349
920	45	9.2	0.1065	13.14	1107.659	96.14872447
940	43	9.4	0.1019	13.42	1111.241	91.69928035
960	38	9.6	0.0904	13.71	1114.975	81.07805108
980	34	9.8	0.0812	14	1118.735	72.58197875



From graph, UCS = 98.553 kN/m²

Our soil type is clay. So, Φ = 0

C = 98.553/2

C = 49.276 kN/m²

Now, lets find Bearing Capacity:-

Square footing = 1m x 1m

Depth, H = 0.5m

$\Phi = 0$ C = 49.276 KN/m² r = 17.365 KN/m³

first assume general shear failure will occur.

Now,

Ultimate Bearing Capacity, $q_u = CNcSc + QNqSq + \frac{1}{2} B\boldsymbol{\gamma N\gamma S\gamma}$.

Where Q = γh

Values of constant from table 1 (IS: 6403)

Nc = 5.14

Nq = 1

Nγ = 0.00

And shape factor (from table 2)

Sc = 1.3

Sq = 1.2

Sγ = 0.8

Now, q_u = 340 KN/m²

And then, Net Bearing Capacity, q_d = 330 KN/m²

Take FOS = 3

Safe Bearing Capacity = 330/3 + **γh**

= **118.68** KN/m²

For given footing allowable load = 110 KN/m²

CONCULSION:

For our square footing size 1m x 1m and depth 0.5m safe bearing capacity

Is 118.68 KN/m² for this footing we can allow 110 KN/m² load for construction